

STATEMENT OF WORK (SOW)

ELECTRICAL INSTALLATION WORKS – RECREATION CENTER RENOVATION PROJECT- PHASE III

1. PROJECT DESCRIPTION

This project comprises the **modification, supply, installation, wiring, testing, and commissioning of the electrical power and lighting systems** for the following areas:

- Clubhouse
- Tool Store
- Security Office / Guard House
- Generator Room
- Associated external and service areas

The work includes **complete system integration**, including **lighting, power distribution, control systems, and associated infrastructure**.

2. GENERAL TECHNICAL REQUIREMENTS (MANDATORY – FROM BOQ NOTES)

The Contractor shall strictly comply with the following:

2.1 Cable Management & Containment

- All cable trays/trunking shall be:
 - Zn-Al powder coated
 - Minimum coating thickness: **100 microns**
 - Only **factory-fabricated fittings** shall be used
 - Site-modified fittings are **strictly prohibited**
 - Provide **HDG (Hot Dip Galvanized) support systems**
 - All tray/trunking joints shall be bonded using:
 - **2.5 sq.mm Cu/PVC earth conductor**
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2.2 Wiring System

- All conduits and accessories shall be:
 - **Low Smoke Zero Halogen (LSZH)**
 - All cables:
 - Minimum insulation rating: **500/750V**
 - Provide:
 - **Proper color coding and identification labeling system**
 - Voltage drop:
 - Must comply with **latest wiring regulations / NEC limits**
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2.3 Compliance & Certification

- Contractor shall:
 - Arrange **third-party inspection and certification**
 - Certification shall be by **licensed/authority-approved electrical engineer**
 - Panel sizing:
 - Must be **coordinated with actual site space constraints**
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3. ELECTRICAL DISTRIBUTION SYSTEM

3.1 Panel Boards (Supply / Modification)

The Contractor shall **supply, install, and modify** the following panels:

Locations & Scope

- **Clubhouse**
 - Modify: **SDB-CL**
- **Clubhouse (AC System)**
 - New: **SDB-AC (VRV/VRF Power Panel)**
- **Tool Store Area**
 - Modify: **MDB-TL**

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- **Security Office**
 - Modify: **SDB-SC**

Scope Includes

- Additional breakers
 - Internal wiring modifications
 - Busbars, terminals, accessories
 - Complete compliance with schematic diagrams
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3.2 Main Power Cabling

Cable Runs

- MDB-TL → SDB-CL
- MDB-TL → SDB-AC

Cable Specification

- 4C x 16 sq.mm Cu/XLPE/LSZH + 16 sq.mm Earth

Scope Includes

- Supply, laying, termination
 - Glands, lugs, sleeves
 - Identification tagging
 - Installation via tray/conduit
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4. ELECTRICAL FITTINGS & LIGHTING SYSTEM

4.1 General Requirements

- All lighting fixtures:
 - Supplied with **drivers and controllers**
 - Driver enclosures included in rate
- Installation shall include:
 - Brackets, supports, chains, threaded rods

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- **Architect approval required** prior to installation (mandatory)
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4.2 Lighting Control Compatibility

All fixtures and drivers shall:

- Be **DALI-2 certified OR KNX compatible**
 - Fully integrate with:
 - **Lutron Athena / KNX system**
 - Support:
 - Digital addressing
 - Flicker-free dimming
 - Status monitoring
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4.3 Lighting Fixtures (Exact BOQ Types)

Provide complete supply and installation of:

- DL1 – recessed LED downlight (16W, 3000K, IP44)
- DL2 – recessed LED downlight (11W, IP20)
- TB1 – weatherproof luminaire (IP66)
- SP1 – RGBW spotlight (DMX512)
- LS1 – suspended LED strip system with aluminum frame
- WL1 – wall light (IP65)
- ST1 – step light (IP65)

(All fixtures include full accessories, wiring, supports, and drivers)

4.4 Socket Outlets & Accessories

General

- All 13A sockets in kitchen/washroom:
 - **IP65 with flip cover**

Items Included

- Floor boxes (OBO Bettermann or equivalent)
- Industrial sockets (16A, IP66, interlocked)
- Standard 13A switched sockets (BS 1363 compliant)

Scope Includes

- Back boxes
 - Plates
 - Internal wiring
 - Fixing accessories
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4.5 Isolators & Devices

- 40A, 4P isolator
 - 63A, 4P isolator
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4.6 Sensors

- Ceiling-mounted motion/presence sensors:
 - 360° coverage
 - 8m detection radius
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5. WIRING INSTALLATION (CRITICAL – FULL BOQ COVERAGE)

5.1 General Wiring Scope

Rates shall include **ALL components required for a complete installation**, including:

- Cables

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- Switches
 - Junction boxes
 - Outlet boxes
 - LSZH conduits
 - Pull boxes
 - Clips, supports
 - Earth wires
 - Draw wires
 - Screws, fixing hardware
 - Cutting, chasing, making good
 - Ceiling boxes and flexible cords
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5.2 Circuit Wiring

Lighting

- 2 x 1C 1.5 sq.mm Cu/XLPE/LSZH + Earth

13A Socket Circuits

- 2C x 2.5 sq.mm + Earth

Industrial Socket Circuits

- Same as above (2.5 sq.mm)

Floor Box Circuits

- 2C x 2.5 sq.mm + Earth
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6. LIGHTING CONTROL SYSTEM (LUTRON / KNX)

Contractor shall:

- Supply, install, connect, test, program, and commission:
 - Lighting control panels
 - Keypads

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- Actuators
- Control cabling

Scope Includes

- Full system wiring (same inclusions as power wiring)
 - Integration with all lighting fixtures
 - Final commissioning and programming
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7. FEEDER CABLES

- 600/1000V rated Cu/XLPE/LSZH cables
- Installed via tray or LSZH conduit

Includes:

- Termination at both ends
 - Glands, lugs, sleeves
 - Tray installation
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8. CABLE TRAYS / TRUNKING SYSTEMS

- Hot-dip galvanized perforated trays/ladders
- Includes:
 - Bends, tees, clamps
 - Anchor fasteners
 - Structural supports at regular intervals

Special Installation:

- 100x40mm trunking for lighting system
 - Must be compatible with lighting fixtures
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9. TESTING & COMMISSIONING

- Full system testing:

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- Insulation resistance
 - Continuity
 - Functional testing
 - Final commissioning of:
 - Power system
 - Lighting
 - Control system
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10. PRELIMINARIES

Contractor shall include:

- Shop drawings
 - Material submittals & approvals
 - Method statements
 - As-built drawings
 - O&M manuals
 - Insurance
 - Bank guarantees
 - Material storage
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11. BUILDER'S WORKS (MANDATORY INCLUSIONS)

Contractor shall allow for:

- Openings in slabs/walls
 - Chasing
 - Masonry works
 - Making good finishes
 - Coordination with all trades
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12. AREA-SPECIFIC WORKS (CRITICAL – FULLY INCLUDED)

The Contractor shall execute **complete electrical installations** for:

12.1 Generator Room

- Cabling
- Distribution boards
- Lighting
- Power outlets
- Fans and associated services

12.2 Tool Room

- Same full scope as above

12.3 Guard House / Security Room

- Same full scope as above

12.4 General

- All associated works required to fully complete these areas
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13. FINAL DELIVERABLE

The Contractor shall provide a **fully operational, tested, certified electrical system**, including:

- Integrated lighting and control system
- Fully functional power distribution
- Complete installation across all specified areas
- Vendor shall do any other related work to meet the functional electrical system in the full phase of renovation and any changes within the 5% of the total cost to be borne by the vendor